Practice Test: AP Precalc 1.1–1.6

Name: _____ Date: _____

About 45 minutes. No calculator unless marked.

1. (1.1) Use the table.

x	-2	-1	0	1	2	3
$f(x)$	8	3	0	-1	0	3

- (a) On what interval is f decreasing? Increasing?
- (b) Where are the zeros?
- (c) Is f concave up or concave down? Justify using the table.

2. (1.2) Find the average rate of change of $f(x) = x^3$ on $[1, 3]$.

3. (1.2) The temperature is $50°$F at 2 PM and $62°$F at 6 PM. Find the average rate of change with units, and say what it means.

4. (1.2) Estimate the rate of change of $g(x) = x^2 + 1$ at $x = 2$ using the intervals $[2, 2.1]$ and $[1.9, 2]$. (Calculator OK)

5. (1.3) Linear, quadratic or neither? Justify.

x	0	1	2	3	4
(a)	1	4	7	10	13
(b)	3	4	7	12	19
(c)	1	2	4	8	16

6. (1.3) $f(x) = 2x^2$. Find the average rate of change on $[0, 1]$, $[1, 2]$ and $[2, 3]$. By how much does it change each time?

7. (1.4) $f(x) = x^3 - 6x^2 + 11x - 6 = (x-1)(x-2)(x-3)$.

- (a) What are the real zeros?
- (b) Is f positive or negative on $(1, 2)$? On $(2, 3)$?
- (c) What is the most turning points f could have?

8. (1.4) A polynomial has degree 5. What is the most real zeros it can have? The most turning points? The fewest real zeros?

9. (1.5) Find all zeros of $f(x) = x^4 - 16$.

10. (1.5) Solve $x^2 + 2x + 5 = 0$.

11. (1.5) Write a polynomial of least degree, with real coefficients and leading coefficient 1, that has a zero of 3 with multiplicity 2 and a zero of i. Give it in standard form.

12. (1.5) For $f(x) = x^2(x-4)^3$, list each zero with its multiplicity and say whether the graph crosses or bounces there.

13. (1.5) Even, odd or neither? Show the test. (a) $x^5 - 2x^3$ (b) $x^4 + x^2 - 7$ (c) $x^2 + x$

14. (1.6) Describe the end behavior with limit notation. (a) $f(x) = 4x^6 - x^7 + 1$ (b) $g(x) = -(x-1)^2(x+3)^2$ (c) $h(x) = 2x(x-1)(x+2)^2$

15. (1.4–1.6) Sketch $f(x) = (x+1)^2(x-2)$. Label the zeros, the y-intercept and the end behavior.

16. (AP-style free response, 1.1–1.4) The water level $W(t)$, in inches, in a tank is measured every hour.

t (hours)	0	1	2	3	4	5
$W(t)$ (in.)	20	25	36	47	52	45

- (a) Find the average rate of change of W from $t = 1$ to $t = 4$. Include units and interpret it.
- (b) Over which one-hour interval(s) does the water level rise fastest?
- (c) Is W concave up or concave down on $[3, 5]$? Justify with the table.
- (d) Based on the table, around when does the water level reach a local maximum? Explain.

Answer Key

1. (a) Decreasing on $(-2, 1)$, increasing on $(1, 3)$. (b) $x = 0$ and $x = 2$. (c) Concave up: the differences −5, −3, −1, 1, 3 are increasing. (It's $x^2 - 2x$.)
2. $\dfrac{27 - 1}{3 - 1} = 13$
3. $\dfrac{62 - 50}{6 - 2} = 3$ °F per hour. On average, the temperature rose 3 degrees each hour between 2 and 6 PM.
4. $\dfrac{5.41 - 5}{0.1} = 4.1$ and $\dfrac{5 - 4.61}{0.1} = 3.9$, so about **4**.
5. (a) Linear: the first differences are all 3. (b) Quadratic: first differences 1, 3, 5, 7; second differences all 2. (c) Neither: the differences 1, 2, 4, 8 aren't constant, and neither are the second differences. (It's exponential, with ratios all 2.)
6. 2, 6, 10. The rate goes up by 4 each time.
7. (a) 1, 2, 3. (b) Positive on $(1, 2)$ (test $x = 1.5$: $(0.5)(-0.5)(-1.5) > 0$). Negative on $(2, 3)$. (c) 2.
8. At most 5 real zeros and at most 4 turning points. At least 1 real zero (odd degree, so the ends go opposite ways and the graph must cross).
9. $(x - 2)(x + 2)(x^2 + 4)$, so $x = \pm 2$ and $\pm 2i$
10. $x = \dfrac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i$
11. The zeros are 3, 3, i and $-i$ (conjugate). $(x - 3)^2(x^2 + 1) = x^4 - 6x^3 + 10x^2 - 6x + 9$
12. $x = 0$, multiplicity 2, bounces. $x = 4$, multiplicity 3, crosses (flattening as it goes through).
13. (a) Odd: $(-x)^5 - 2(-x)^3 = -x^5 + 2x^3 = -f(x)$. (b) Even: all even powers, so $f(-x) = f(x)$. (c) Neither: $f(-x) = x^2 - x$, which is neither $f(x)$ nor $-f(x)$.
14. (a) $\lim_{x \to -\infty} f = \infty$ and $\lim_{x \to \infty} f = -\infty$. (b) Leading term $-x^4$: $\lim_{x \to \pm\infty} g = -\infty$. (c) Leading term $2x^4$: $\lim_{x \to \pm\infty} h = \infty$.
15. Zeros: −1 (bounces) and 2 (crosses). y-intercept $(1)^2(-2) = -2$. Degree 3, positive leading coefficient: down on the left, up on the right. The graph comes up from the bottom left, touches the axis at −1, turns down through $(0, -2)$, then turns back up and crosses at 2.
16. (a) $\dfrac{52 - 25}{4 - 1} = 9$ inches per hour. Between hour 1 and hour 4, the water rose 9 inches per hour on average. (b) $[1, 2]$ and $[2, 3]$, both 11 inches per hour. (c) Concave down: the one-hour changes are 5 then −7, so the rate of change is decreasing. (d) Around $t = 4$: the level rises up to $t = 4$ (change +5) and then falls (change −7), so it switches from increasing to decreasing there.